Math 50

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Unit 3

The CLT, uncertainty quantification, and single-predictor regression

In this unit, we address the question of statistical error. We begin with the **Law of Large Numbers** and **Central Limit Theorem**, which connects the behavior of large random samples to the Normal distribution from Unit 2. We then define an **estimator** (something we have already seen examples of) and discuss the concepts of **bias**, **variance**, **confidence intervals**, and the sample distribution. We apply these ideas to the **least squares estimator** for the regression coefficient in a linear regression model and autoregressive models. Finally, we discuss **hypothesis testing** briefly.

CONCEPTS

The law of large numbers, the Central Limit Theorem, estimators, sample distributions, bias and consistency, standard errors, confidence intervals, the bias-variance decomposition, fitting the single-predictor linear regression model via least squares, autoregressive models and the Hurwicz bias, and hypothesis testing and p-values for regression models.

THINGS TO PRACTICE

- Use the CLT to approximate the distribution of a sum or sample average, and to justify treating an estimator's sample distribution as approximately Normal.
- Identify whether a simple estimator is biased or not, by hand if possible, or using simulations in Python.
- Know the properties of the least squares estimator and what its sample distribution looks like.

- Work with AR(1) models: find the stationary mean and variance, compute autocorrelations, and explain why the least squares estimator is biased.
- Perform a hypothesis test by simulating the model.

3.1 The Law of Large Numbers & the Central Limit Theorem

The goal of this section is to understand the distribution of a sum of iid random variables when N is large — this matters if we want to be precise about how accurate our estimates are, which is the whole subject of this unit. We begin with the **law of large numbers**, which makes precise the statement that a sample average approximates the expectation.

The Law of Large Numbers

The binomial distribution illustrates a basic principle we've already used a number of times: when we sum many independent random variables and divide by the total number, the result lands close to the mean.

Theorem (Law of Large Numbers). Let X_i be iid and $S_N = \sum_{i=1}^{N} X_i$. If $E[X_i] < \infty$, then $S_N/N \to E[X_i]$.

This isn't fully precise — we'd need to define what it means for a random quantity to "converge" — but for our purposes it suffices to think of this as: for large enough N, S_N/N won't differ from $E[X_i]$ by much. Another way to say it: for iid X_i, the sample average $\overline{X}$ approaches $E[X_i]$.

The binomial distribution tells us more: the variation around $E[X_i]$ is proportional to $1/\sqrt{N}$. It's natural to ask whether this holds for other random variables too. The key insight is that the N-dependence in $E[Y] = qN, \ \operatorname{var}(Y) = Nq(1-q)$ doesn't actually depend on the underlying distribution being Bernoulli — so if X_i is a die roll, or a uniform random variable, we'd expect the same scaling to hold. The behavior of random sums is, in

fact, even more universal than this: we can describe the distribution of *any* random sum (with finite variance) using a single distribution — the Normal distribution from Unit 2.

The Central Limit Theorem and sample distribution

We can now state the **Central Limit Theorem** (CLT) precisely.

Theorem. Let X_i be iid with $E[X_i] = \mu$, $\text{var}(X_i) = \sigma^2$, and let $S_N = \sum_{i=1}^{N} X_i$. Then

$$P\left(\frac{S_N - N\mu}{\sqrt{N\sigma^2}} < z\right) \to P(Z < z), \qquad Z \sim \text{Normal}(0, 1).$$

The variable Z is called a **standard normal** random variable. Its CDF is common enough that we abbreviate $\Phi(z) = P(Z < z)$, and write $\phi(z)$ for its pdf. Using the properties of Normal random variables from Unit 2, we can restate the CLT informally as $S_N \approx \text{Normal}(N\mu,\ N\sigma^2)$. In the demo below, the histogram is of standardized sums $(S_N - N\mu)/\sqrt{N\sigma^2}$ and the curve is the standard Normal density: whatever distribution you draw the X_i from, the histogram closes on that curve as N grows.

Note on the iid assumption: the version of the CLT stated above assumes the X_i are independent and identically distributed. There are more advanced versions that allow some dependence, but not arbitrary dependence; strong correlation can change the sample distribution substantially. For this course, check first whether iid is a reasonable approximation before using the CLT.

Drill

Standardizing a sum

Let $X_1, \ldots, X_N$ be iid with $E[X_i] = \mu$ and $\text{var}(X_i) = \sigma^2$, and let $S_N = \sum_i X_i$.

a. Find $E[S_N]$ and $\text{var}(S_N)$.

b. Write the standardized version of S_N.

c. Use the CLT to write an approximate distribution for S_N.

Binomial approximation

Let $Y = \sum_{i=1}^{N} X_i$ where $X_1, \ldots, X_N$ are iid $\text{Bernoulli}(q)$ (this is what it means to say $Y \sim \text{Binomial}(N, q)$).

a. Use the CLT to approximate the distribution of Y.

b. Use that approximation to estimate $P(Y < 70)$ when $N = 200$ and $q = 0.3$.

c. Write Python code using `scipy.stats.norm` and `scipy.stats.binom` to compare the CLT approximation with the exact Binomial probability.

d. Explain why the approximation should improve as N increases.

3.2 Estimators, bias & consistency

We've already alluded to the concept of statistical inference and parameter estimation. Recall that $\hat{q} = \overline{Y}$ is an estimator of the parameter q for a Bernoulli distribution, and more generally we've gone back and forth between "data world" (sample means, histograms) and "math world" (expectation, density) via sample averages.

Statistical inference is the process of estimating the parameters of a distribution (e.g. μ and σ) from samples of Y, *and* expressing our uncertainty in these estimates. The "expressing uncertainty" part is what we haven't yet discussed formally.

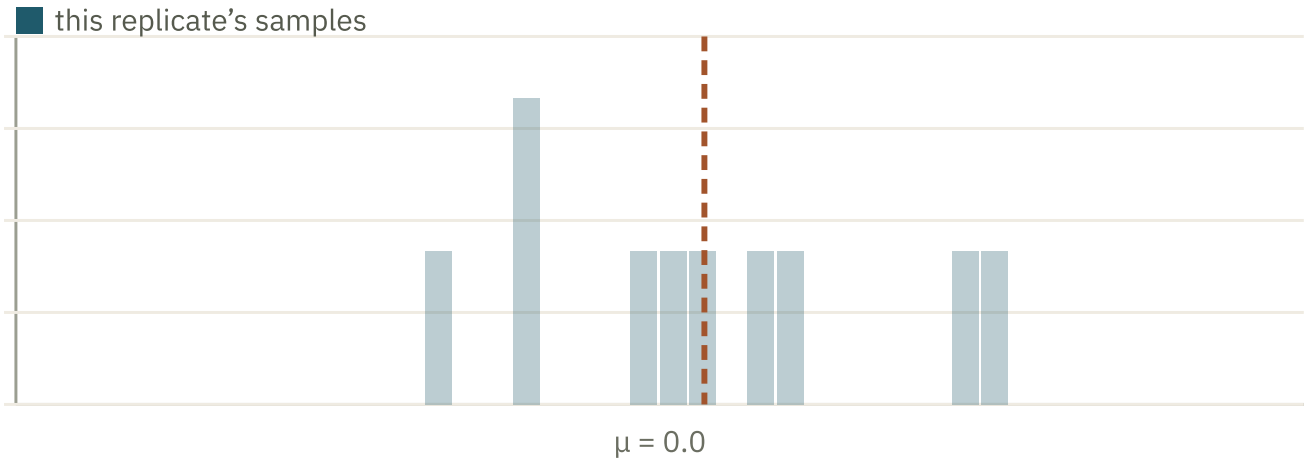
In general, an **estimator** $\hat{\theta}$ of a parameter θ is anything we compute from the data that's meant to approximate θ — in principle this could be any quantity (like the maximum value), but it will almost always take the form of a sample average in some way, as we'll see. The key point: **the estimator is a function of the data** — $\hat{\theta}$ depends on the specific sample or simulation we run, whereas the parameter it approximates does not depend on the data and is (in classical statistics) a fixed number. For instance, in a YES/NO survey or a two-candidate election, the "true" quantity we want to measure is the population fraction who

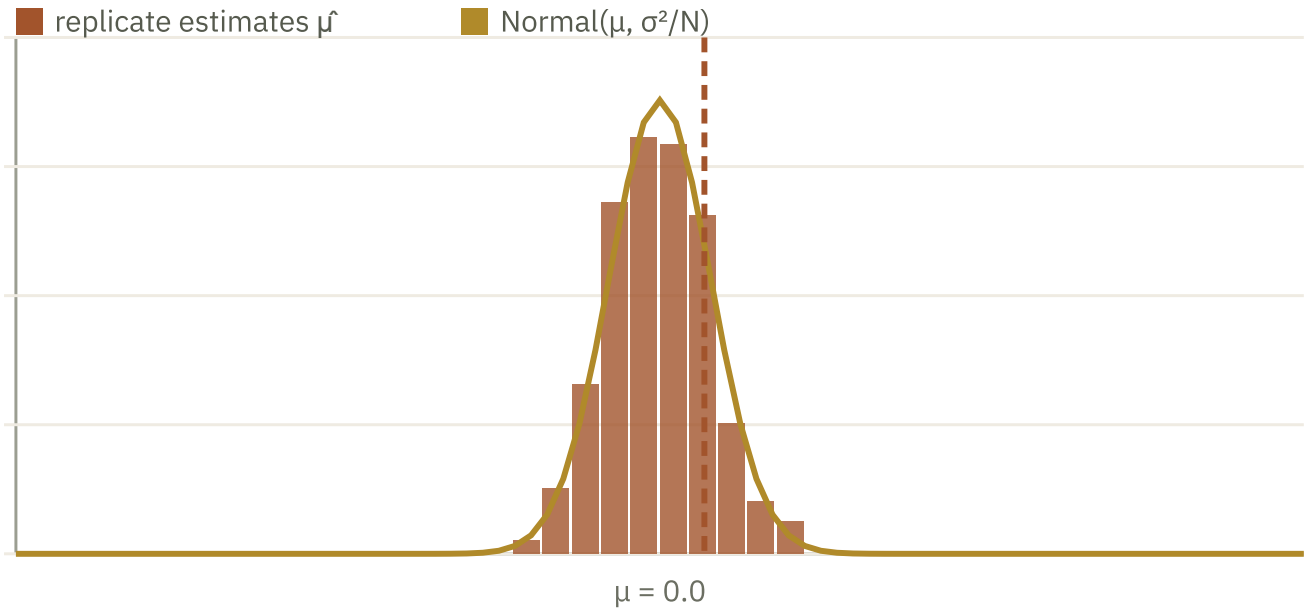
would answer YES. Our estimate $\hat{q}$ is a random variable that depends on which subset of the population we happen to sample, and it changes if we sample a different subset.

Sample distribution and standard errors

We call the distribution of $\hat{\theta}$ over many replications of our data the **sample distribution**. We use **replicate** to mean a different realization of the *entire* dataset (as opposed to the individual samples *within* a dataset) — the terminology is a little confusing: the sample distribution is the distribution of $\hat{\theta}$ over many replicates, but each replicate itself consists of many samples. The demo below makes the distinction concrete: the pale bars (top) are a histogram of the N *samples* Y_i inside the latest replicate, while the solid bars (bottom) collect one estimate $\hat{\mu} = \overline{Y}$ per replicate — that second histogram is the sample distribution, compared against the theoretical $\text{Normal}(\mu, \sigma^2/N)$ curve.

Replicates vs. samples, and the sample distribution of $\hat{\mu}$





$Y \sim \text{Normal}(\mu, \sigma^2)$ with $\mu = 0$, $\sigma = 1.5$ fixed. The pale bars (top) are the N individual samples Y_i from the latest replicate; the solid bars (bottom) are $\hat{\mu}$ collected one per replicate, compared against the theoretical $\text{Normal}(\mu, \sigma^2/N)$ curve.

Example (sample distribution of a Normal mean)

Suppose $Y \sim \text{Normal}(\mu, \sigma^2)$.

Question: what is the sample distribution of $\hat{\mu}$ (our estimator of μ)?

Solution:

$$\hat{\mu} = \overline{Y} = \frac{1}{N} \sum_{i=1}^{N} Y_i.$$

Because sums of independent Normal random variables are exactly Normal,

$$\sum_{i=1}^{N} Y_i \sim \text{Normal}(N\mu,\ N\sigma^2).$$

For non-Normal variables with finite variance, the CLT gives this same statement as a large-sample approximation. Dividing by N,

$$\hat{\mu} \sim \text{Normal}\left(\mu,\ \frac{\sigma^2}{N}\right).$$

This assumes σ is known.

A natural way to quantify our uncertainty in $\hat{\mu}$ is the standard deviation of $\hat{\mu}$ under the sample distribution. We call this the **standard error**: our estimate of the standard deviation of the sample distribution. For the Normal model, if we're estimating the mean and happen to know σ,

$$\text{se}(\hat{\mu}) = \frac{\sigma}{\sqrt{N}}.$$

This tells us how much our estimate would vary between different experiments (or surveys, or simulations). Importantly, the standard error depends on σ, which we usually *don't* know! So it's common to plug in an estimate $\hat{\sigma}$ instead:

$$\text{se}(\hat{\mu}) = \frac{\hat{\sigma}}{\sqrt{N}}.$$

It should be clear from context which one is meant: if we're working with real data and don't know σ, "standard error" means the second formula; if we're working with a model where the parameters are specified, it means the first.

Bias and consistency

There must be some minimal properties we'd like an estimator to have. At the very least, it should be informed by the data in the sense that more data brings $\hat{\theta}$ closer to the true θ — more precisely, larger N should make $\hat{\theta}$ closer to θ. Of course we need to say what "closer" means for random quantities. We'll say $\hat{\theta}$ is **consistent** if

$$E[\hat{\theta}] \to \theta \quad \text{and} \quad \text{se}(\hat{\theta}) \to 0 \quad \text{as } N \to \infty,$$

i.e. as we collect more data, the sample distribution becomes more concentrated around θ.

Consistency alone isn't the whole story: $\hat{\mu}_1 = \hat{\mu} + 1/N$ is also consistent, yet clearly worse than $\hat{\mu}$. To rule this out we say $\hat{\theta}$ is **unbiased** if, for *every* N (not just large N), the average over the sample distribution equals the true value:

$$E[\hat{\theta}] = \theta.$$

Example (bias and consistency)

For a Normal random variable, define the estimator of the mean

$$\hat{\mu}_2 = \frac{Y_1 + Y_2}{2}.$$

Question: is $\hat{\mu}_2$ biased? Is it consistent? What's its sample distribution?

Solution: $\hat{\mu}_2$ has sample distribution

$$\hat{\mu}_2 \sim \text{Normal}\left(\mu,\ \frac{\sigma^2}{2}\right).$$

It is therefore unbiased, but *not* consistent, since $\text{se}(\hat{\mu}_2) = \sigma/\sqrt{2}$ doesn't shrink as we collect more data (it only ever uses two samples).

Example (Bessel's correction)

Now consider estimating the variance of $Y \sim \text{Normal}(\mu, \sigma^2)$. Given samples $Y_1, \ldots, Y_N$, the natural way to estimate σ^2 is

$$\text{var}(Y) = E\big[(Y - E[Y])^2\big] \approx \hat{\sigma}_0^2 := \frac{1}{N}\sum_{i=1}^{N}(Y_i - \overline{Y})^2.$$

It turns out $\hat{\sigma}_0^2$ is *biased* — in fact

$$\hat{\sigma}^2 := \frac{1}{N-1}\sum_{i=1}^{N}(Y_i - \overline{Y})^2 = \frac{N}{N-1}\hat{\sigma}_0^2$$

is unbiased. The correction factor $N/(N-1)$ is called **Bessel's correction**.

Question: confirm with simulated data that $\hat{\sigma}_0^2$ is biased while $\hat{\sigma}^2$ is not.

Bias-variance tradeoff

Define the **mean-squared error** of an estimator $\hat{\theta}$ of a quantity θ (which could be a parameter, or the value of a function like $f(x)$ we're trying to predict):

$$\text{MSE}_{\hat{\theta}} = E\big[(\hat{\theta} - \theta)^2\big].$$

The following is the key result behind the **bias-variance tradeoff** — the U-shaped tradeoff curves we'll see later in the course.

Theorem (bias-variance decomposition).

$$\text{MSE}_{\hat{\theta}} = \text{var}(\hat{\theta}) + \big(E[\hat{\theta}] - \theta\big)^2.$$

▼ **Proof**

Using the definition of variance,

$$\text{var}(\hat{\theta} - \theta) = E\big[(\hat{\theta} - \theta)^2\big] - \big(E[\hat{\theta} - \theta]\big)^2.$$

Since θ is a constant, $\text{var}(\hat{\theta} - \theta) = \text{var}(\hat{\theta})$; rearranging gives the result.

An estimator's error therefore decomposes into how *variable* it is (variance) plus how far off it is *on average* (squared bias).

Aside: maximum likelihood estimation (optional)

Sometimes the "right" estimator for a parameter is obvious — that's the case for q in the Bernoulli distribution. But this won't always be true, so it's useful to have a more systematic way of finding estimators.

Recall the Binomial pmf,

$$P(Y) = \binom{N}{Y} q^Y (1-q)^{N-Y}.$$

In statistics we sometimes call this the **likelihood**, written $L(Y \mid q)$ — the notation suggests we're thinking of P as a distribution conditioned on a particular parameter value. More generally, the likelihood is the probability of the observed data given the parameters; this notation foreshadows Bayesian thinking, where the parameters themselves are treated as random variables (more on that later).

The equation above tells us how likely it is to observe Y YESes among N people surveyed. It seems reasonable that this shouldn't be too small, since that would mean our results are an anomaly — the larger $L(Y \mid q)$ is, the more "likely" our data. This suggests estimating q by the value $\hat{q}$ that makes $L(Y \mid q)$ largest: the **maximum likelihood estimate** (MLE). Using calculus (try it!), the maximizer is

$$\hat{q}_{\text{MLE}} = \frac{Y}{N}.$$

For a Normal distribution with mean μ and variance σ^2, the MLEs are the usual sample mean and (biased) sample variance we've already seen.

Drill

Bias and consistency

Let $X \sim \text{Bernoulli}(q)$ and $X_1, \ldots, X_N$ denote N samples of X. For each estimator of q, write down the standard error and state whether it is unbiased and/or consistent. You can use exact variance formulas; you do not need the CLT.

a. $$\hat{q}_0 = \frac{1}{N} \sum_{i=1}^{N} X_i$$

b. $$\hat{q}_1 = \frac{Y}{N} + \frac{1}{\sqrt{N}}, \qquad Y = \sum_{i=1}^{N} X_i$$

c. $$\hat{q}_2 = \frac{1}{\lfloor N/2 \rfloor} \sum_{i=1}^{\lfloor N/2 \rfloor} X_i$$

The notation $\lfloor n \rfloor$ means the largest integer $\leq n$. For example, $\lfloor 101/2 \rfloor = 50$.

Comparing two estimators

Let $X_1, \ldots, X_N$ be iid with $E[X_i] = \mu$ and $\text{var}(X_i) = \sigma^2$, and consider

$$\hat{\mu}_A = \overline{X}, \qquad \hat{\mu}_B = \frac{1}{N} \sum_{i=1}^{N} X_i + \frac{2}{N}.$$

a. Find the bias and standard error of each estimator.

b. Which estimators are unbiased? Which are consistent?

c. Compare their mean-squared errors. Which estimator would you prefer for finite N?

Bessel's correction

For $N = 2$ iid samples Y_1, Y_2 with mean μ and variance σ^2,

$$\hat{\sigma}_0^2 = \frac{1}{2}\left[(Y_1 - \overline{Y})^2 + (Y_2 - \overline{Y})^2\right], \qquad \overline{Y} = \frac{Y_1 + Y_2}{2}.$$

a. Show that $\hat{\sigma}_0^2 = \frac{1}{4}(Y_1 - Y_2)^2$.

b. Compute $E[\hat{\sigma}_0^2]$ in terms of σ^2.

c. Confirm this matches $\frac{N-1}{N}\sigma^2$ for $N = 2$.

Bias, variance and MSE from a table

An estimator $\hat{\theta}$ of $\theta = 2$ has the sample distribution below.

$\hat{\theta}$	1	2	4
probability	0.25	0.5	0.25

a. Compute the bias of $\hat{\theta}$.

b. Compute $\mathrm{var}(\hat{\theta})$.

c. Compute $\mathrm{MSE}_{\hat{\theta}} = E[(\hat{\theta} - \theta)^2]$ directly from the table, and check that it equals $\mathrm{var}(\hat{\theta}) + \mathrm{bias}^2$.

A weighted average

Let Y_1, Y_2 be independent with $E[Y_i] = \mu$ and $\mathrm{var}(Y_i) = \sigma^2$, and for a number $0 \le w \le 1$ let

$$\hat{\mu}_w = wY_1 + (1-w)Y_2.$$

a. Show that $\hat{\mu}_w$ is unbiased for every w.

b. Find $\mathrm{var}(\hat{\mu}_w)$. Which w makes it smallest, and what is the smallest value?

Shrinking the sample mean

Let $Y_1, \dots, Y_N \sim \mathrm{Normal}(\mu, \sigma^2)$ and, for a fixed number $a \ge 0$, consider

$$\hat{\mu}_a = \frac{\overline{Y}}{1+a}.$$

a. Find the bias and the variance of $\hat{\mu}_a$. What is its sample distribution?

b. Write down $\mathrm{MSE}_{\hat{\mu}_a}$. Is $\hat{\mu}_a$ consistent when $a > 0$?

c. Suppose the true mean is small compared with the standard error, $\mu^2 \le \sigma^2/N$. Show that $\mathrm{MSE}_{\hat{\mu}_a} < \mathrm{MSE}_{\overline{Y}}$ for every $a > 0$. What does this say about whether unbiased estimators are always best?

Estimating μ^2

Let $Y_1, \dots, Y_N \sim \mathrm{Normal}(\mu, \sigma^2)$. To estimate μ^2 we use $\overline{Y}^2$.

a. Compute $E[\overline{Y}^2]$.

b. Is $\overline{Y}^2$ an unbiased estimator of μ^2? Is it consistent?

c. Even though $\overline{Y}$ is unbiased for μ, $\overline{Y}^2$ is not unbiased for μ^2. In one sentence, why not?

Estimating q from the variance

Let $Y_1, \dots, Y_N \sim \mathrm{Bernoulli}(q)$ and suppose we know $q < 1/2$. Besides $\hat{q} = \overline{Y}$, we could estimate the variance with $\hat{\sigma}_0^2 = \frac{1}{N}\sum_i (Y_i - \overline{Y})^2$ and then solve $\hat{\sigma}_0^2 = q(1-q)$ for q.

a. Show that $\hat{\sigma}_0^2 = \overline{Y}(1 - \overline{Y})$. (Hint: each Y_i is 0 or 1.)

b. Solve $v = q(1-q)$ for q. There are two solutions; which one should we use, and why do we need to know $q < 1/2$?

c. Plug in $v = \hat{\sigma}_0^2$ and show that the new estimator is $\min(\overline{Y},\, 1 - \overline{Y})$.

d. Is this estimator biased?

3.3 Confidence intervals

The idea of a **confidence interval** is, roughly, to describe the range of values where the true θ plausibly lies, given an estimate $\hat{\theta}$ and its sample distribution. We'll mostly use the 95% confidence interval (95%-CI),

$$\left[\hat{\theta} - 1.96\,\mathrm{se}(\hat{\theta}),\ \ \hat{\theta} + 1.96\,\mathrm{se}(\hat{\theta})\right].$$

For an approximately Normal sample distribution, 95% of values lie within 1.96 standard errors of its center; this is why the resulting intervals have about 95% coverage.

When the Normal approximation is not quite right: the t distribution. In practice we plug an estimate $\hat{\sigma}$ into the standard error. For Normal data, $(\hat{\mu} - \mu)/\mathrm{se}(\hat{\mu})$ with $\mathrm{se}(\hat{\mu}) = \hat{\sigma}/\sqrt{N}$ is then not exactly $\mathrm{Normal}(0, 1)$: it has a t **distribution**, which looks like a Normal but has heavier tails because $\hat{\sigma}$ is itself random; how much heavier depends on N. The 95% multiplier is then a bit larger than 1.96: about 2.78 for $N = 5$, 2.26 for $N = 10$, 2.05 for $N = 30$. For $N \gtrsim 30$ the difference is negligible, which is why we use 1.96 throughout; for small N, use the larger t multiplier. Software such as `statsmodels` reports t-based p-values and intervals, so its numbers may differ slightly from the Normal ones.

Note that samples from *this* interval do not have the same distribution as $\hat{\theta}$ over replicates of our data. Put another way: if we draw many samples from our estimated sample distribution, their spread is not the same as the spread of $\hat{\theta}$ we'd get by actually running the experiment many times. The correct interpretation of the 95%-CI is: **if we generate many replicates of the data, the true value θ will fall inside the CI for 95% of them.**

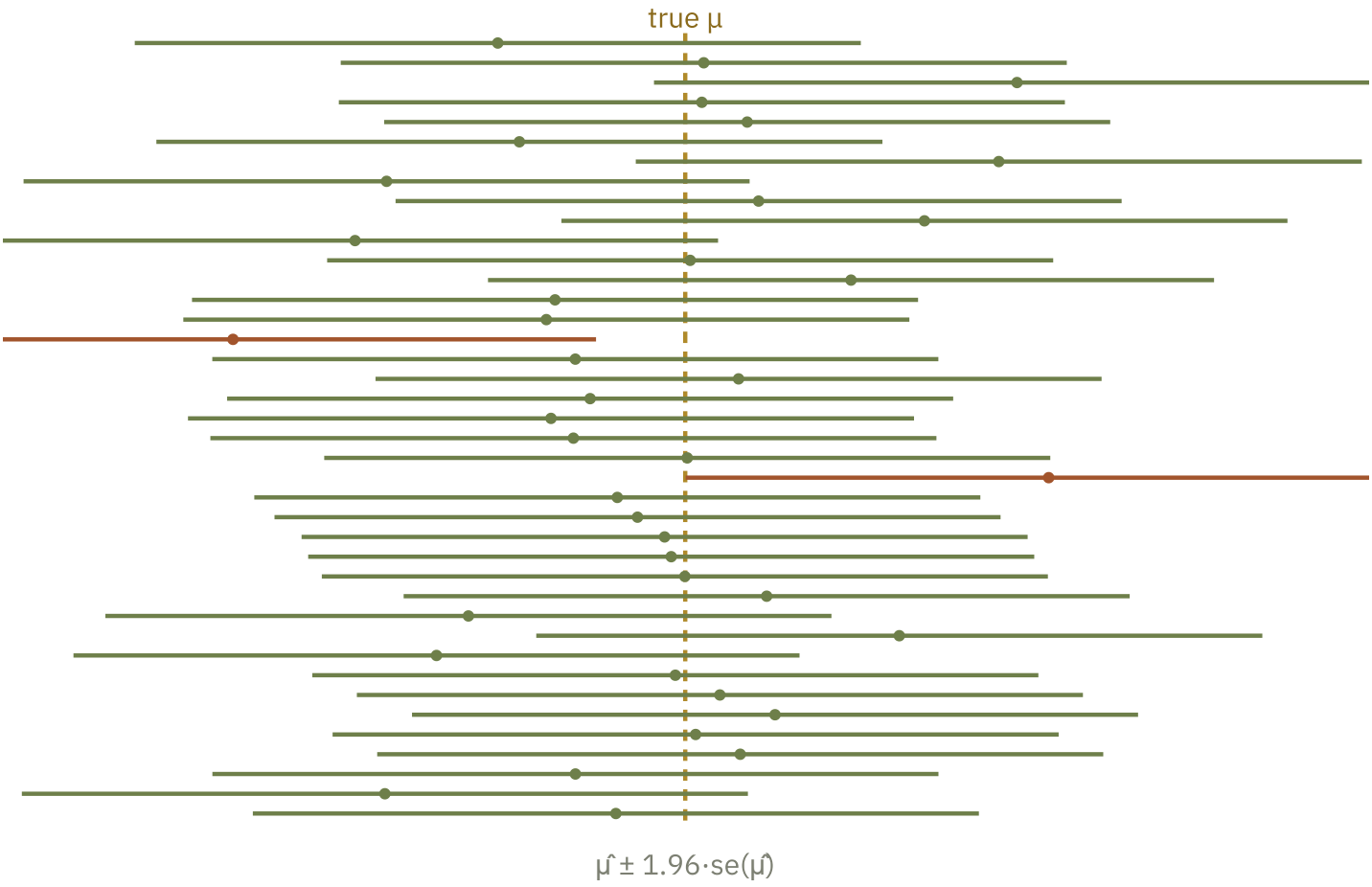
Technically, it is *not* true that there's a 95% chance the true value lies in the particular 95%-CI computed from one dataset. To see why, note that θ has a 95% chance of lying in

$$\left[\theta - 1.96\,\mathrm{std}(\hat{\theta}),\ \ \theta + 1.96\,\mathrm{std}(\hat{\theta})\right],$$

but this is different from the interval above, since here we've replaced $\hat{\theta}$ with θ. The distinction is illustrated in the demo below, where each horizontal bar is the 95%-CI from

one replicate and is coloured by whether it contains the true value — every CI is centered at a different (random) $\hat{\theta}$, while the dashed line marking θ stays fixed. It's a subtle point, and for practical purposes you can mostly think of the 95%-CI as "the region where the parameter is likely to be." (We'll revisit this with a genuinely different, Bayesian, interpretation later in the course.)

95%-CI



Example (sample size for a target CI width)

Suppose we're designing an experiment. Our model is $Y_1, \ldots, Y_N \sim \text{Normal}(\mu, \sigma^2)$, we will estimate μ with $\hat{\mu} = \overline{Y}$, and from previous experiments we have the estimate $\sigma \approx 1$.

Question: how large must N be so the 95% confidence interval, $\hat{\mu} \pm 1.96\,\mathrm{se}(\hat{\mu})$, is no wider than $\hat{\mu} \pm 0.25$?

Solution: the standard error based on N samples is $\sigma/\sqrt{N} \approx 1/\sqrt{N}$, so we need

$$\frac{1.96}{\sqrt{N}} \le 0.25 \implies \sqrt{N} \ge 7.84 \implies N \ge (7.84)^2 \approx 61.5,$$

so $N = 62$. Since $\sigma \approx 1$ is itself only an estimate, this is approximate too. We can check it by running many replicates for each N (as in the class notebook).

Drill

Reading a confidence interval

Suppose an estimator has $\hat{\theta} = 4.8$ and $\mathrm{se}(\hat{\theta}) = 0.7$.

a. Compute the approximate 95% confidence interval.

b. Explain what "95% confidence" means in terms of repeated datasets.

c. Explain why it is not literally a statement that θ has probability 0.95 of lying in this particular interval.

Target width

For a Normal model with known $\sigma = 2$, find the minimum N so that the approximate 95% confidence interval for μ has total width at most 1.

3.4 Least squares and estimators for linear regression

We now discuss estimators for the parameters of a single-predictor linear regression model. Recall from Unit 2 that a **linear regression model** is

$$Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X,\ \sigma_\epsilon^2), \qquad \text{i.e.} \qquad Y = \beta_0 + \beta_1 X + \epsilon, \quad \epsilon \sim \text{Normal}(0, \sigma_\epsilon^2).$$

The input is the **predictor** (X) and the output is the **response variable** (Y). Our data are N pairs $(X_1, Y_1), \ldots, (X_N, Y_N)$.

Estimating β_0 and β_1

Recall from Unit 2 the **covariance** $\text{cov}(X, Y) = E[XY] - E[X]E[Y]$, and that for any linear regression model, regardless of the distribution of X,

$$\beta_1 = \frac{\text{cov}(X, Y)}{\sigma_X^2}.$$

To estimate the covariance from data, it helps to write it in a different form. Start from $E\big[(X - E[X])(Y - E[Y])\big]$ and simplify:

$$\begin{aligned} E\big[(X - E[X])(Y - E[Y])\big] &= E\big[XY - E[Y]X - E[X]Y + E[X]E[Y]\big] && \text{(expan} \\ &= E[XY] - E[Y]E[X] - E[X]E[Y] + E[X]E[Y] && \text{(linear} \\ &= E[XY] - E[X]E[Y] = \text{cov}(X, Y). \end{aligned}$$

Similarly $\sigma_X^2 = E\big[(X - E[X])^2\big]$. Now replace each expectation with a sample average, $E[\,\cdot\,] \to \frac{1}{N}\sum_i$, and $E[X], E[Y]$ with $\overline{X}, \overline{Y}$:

$$\text{cov}(X, Y) \approx \frac{1}{N}\sum_{i=1}^{N}(X_i - \overline{X})(Y_i - \overline{Y}), \qquad \sigma_X^2 \approx \frac{1}{N}\sum_{i=1}^{N}(X_i - \overline{X})^2.$$

The $\frac{1}{N}$'s cancel in the ratio, and taking expectations of the model, $E[Y] = \beta_0 + \beta_1 E[X]$, gives β_0. So

$$\boxed{\hat{\beta}_1 = \frac{\sum_{i=1}^{N}(X_i - \overline{X})(Y_i - \overline{Y})}{\sum_{i=1}^{N}(X_i - \overline{X})^2}, \qquad \hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X}.}$$

In Python, `np.cov(x,y)[0,1] / np.cov(x,y)[0,0]` computes $\hat{\beta}_1$ directly.

Least squares, SSR and SST

The **residuals** are the vertical distances from each point to the fitted line,

$$r_i = Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i),$$

and we define two sums of squares:

$$\mathrm{SSR} = \sum_{i=1}^{N} r_i^2, \qquad \mathrm{SST} = \sum_{i=1}^{N} (Y_i - \overline{Y})^2.$$

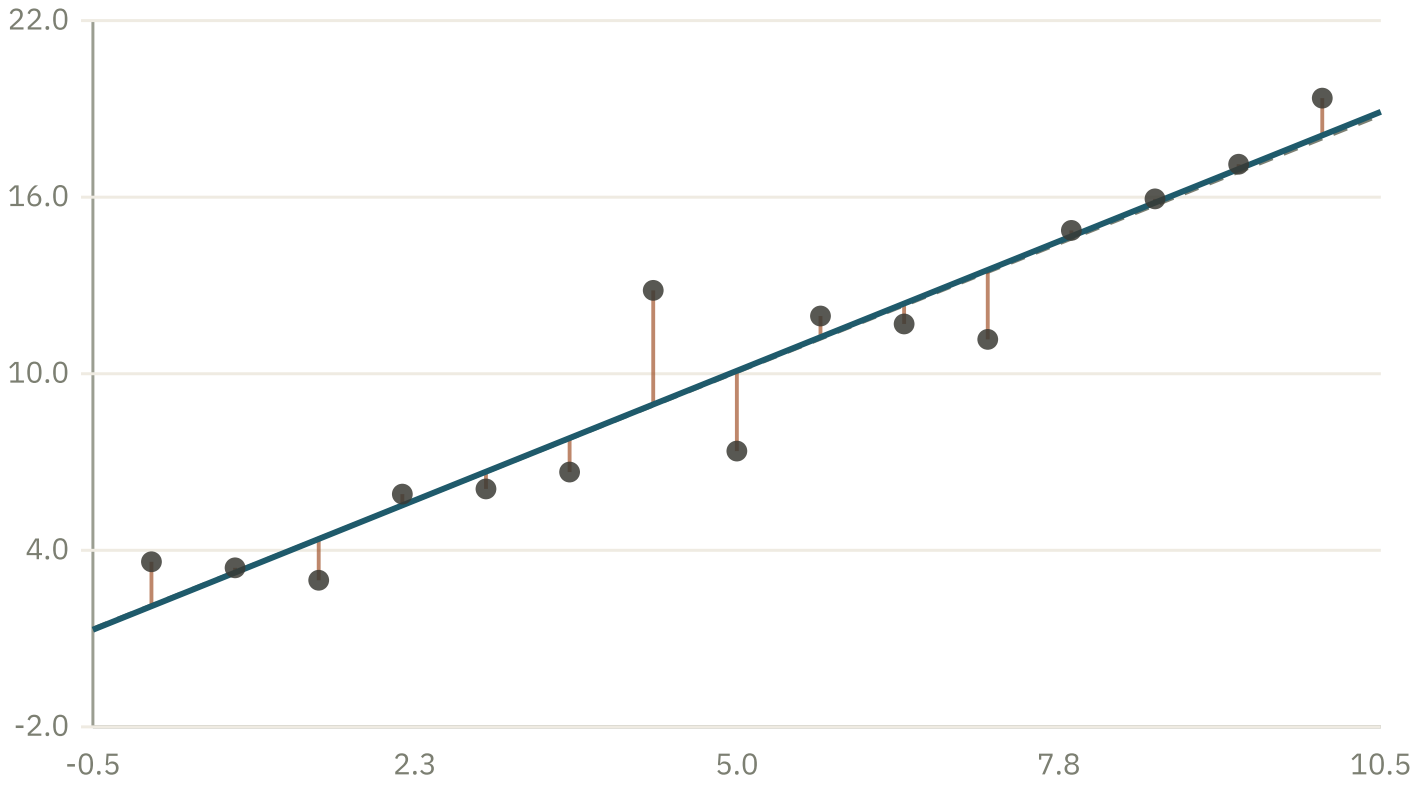
SSR is the **sum of squared residuals**, also called the residual sum of squares (RSS) or the sum of squared errors (SSE). SST is the **total sum of squares**, also written TSS or SS_{tot}. (Be careful when reading other sources: some books use "SSR" for the *regression* sum of squares, $\mathrm{SST} - \mathrm{SSE}$, which is a different quantity.)

Regardless of where the (X, Y) points came from (a Normal model or not), $\hat{\beta}_0$ and $\hat{\beta}_1$ above are exactly the values that give the line with the smallest SSR:

$$(\hat{\beta}_0, \hat{\beta}_1) = \underset{b_0,\, b_1}{\operatorname{argmin}} \sum_{i=1}^{N} \left(Y_i - b_0 - b_1 X_i\right)^2,$$

which is why they are called the **least squares** estimators. There are many other ways we could draw a line through a set of points (for example minimizing $\sum_i |r_i|$, or the perpendicular distances to the line); minimizing SSR happens to be the right choice under the assumption that the data come from a linear regression model. In the demo below the vertical segments are the residuals r_i. Try to fit the line by eye first, then compare against the true minimizer of SSR.

Least squares: minimizing the residual sum of squares



SSR and SST also give us estimators of the remaining quantities in the model:

$$\boxed{\hat{\sigma}_\epsilon^2 = \frac{\text{SSR}}{N-2}}, \qquad \boxed{\hat{\rho}^2 = R^2 = 1 - \frac{\text{SSR}}{\text{SST}}}.$$

We divide by $N-2$ rather than N because two parameters (β_0, β_1) were estimated from the same data and this is needed to make the estimator unbiased. For large N this is not so important and you case just remembder that $\hat{\sigma}_\epsilon^2 \approx \text{SSR}/N$.

Example (marketing data)

The **Advertising data** (from *An Introduction to Statistical Learning*) records, for $N = 200$ markets, the budgets spent on TV, radio and newspaper advertising for a product (in thousands of dollars) and the product's sales (in thousands of units). We'll explore whether the TV budget is associated with higher sales, with X the TV budget and Y the sales:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("https://www.statlearning.com/s/Advertising.csv")
x = df["TV"].to_numpy()
y = df["sales"].to_numpy()
```

Question: fit a linear regression model with the TV budget as the predictor and sales as the response.

a. Compute $\hat{\beta}_0$, $\hat{\beta}_1$, SSR, SST, $\hat{\sigma}_\epsilon$ and R^2. Interpret $\hat{\beta}_1$ and R^2.

b. Plot the regression line together with a scatter plot of the data.

c. Using the fitted $\hat{\beta}_0, \hat{\beta}_1, \hat{\sigma}_\epsilon$, simulate 10 "fake" datasets with the same x values and number of points as the real data, and compare their plots to the real data — does the model look reasonable?

Solution: (a) Each line implements one of the formulas above:

```
N = len(x)
beta1_hat = np.sum((x - np.mean(x)) * (y - np.mean(y))) / np.sum((x - np.m
beta0_hat = np.mean(y) - beta1_hat * np.mean(x)

r = y - (beta0_hat + beta1_hat * x)        # residuals
SSR = np.sum(r**2)
SST = np.sum((y - np.mean(y))**2)
sigma_hat = np.sqrt(SSR / (N - 2))
R2 = 1 - SSR / SST
```

This gives $\hat{\beta}_0 \approx 7.03$, $\hat{\beta}_1 \approx 0.048$, $\text{SSR} \approx 2103$, $\text{SST} \approx 5417$, $\hat{\sigma}_\epsilon \approx 3.26$ and $R^2 \approx 0.61$. So each extra $\$1000 of TV advertising is associated with about 48 more units sold, and the TV budget explains about 61\% of the variation in sales between markets (recall \rho^2$ from Unit 2.4; with one predictor, R^2 also equals the squared sample correlation, `np.corrcoef(x, y)[0, 1]**2`).

(b) and (c):

```
xx = np.linspace(0, np.max(x), 100)
plt.scatter(x, y, alpha=0.8, label="real data")
plt.plot(xx, beta0_hat + beta1_hat * xx, color="red", label="fitted line")
for k in range(10):
    y_fake = beta0_hat + beta1_hat * x + np.random.normal(0, sigma_hat, N)
    plt.scatter(x, y_fake, alpha=0.15, color="gray", label="fake data" if
plt.xlabel("TV budget ($1000s)")
plt.ylabel("sales (1000s of units)")
plt.legend()
plt.show()
```

The fake datasets have roughly constant spread around the line, while the real data fan out: sales are much more variable in markets with large TV budgets. So the constant-variance assumption of the model is questionable here: the residual standard deviation grows from about 1.8 for budgets under 100 to about 4.4 for budgets over 200.

Drill

Estimating the regression slope by hand

Estimate the regression slope and intercept of the following dataset by hand:

X	Y
1	2
2	3
3	5
4	6
5	8

Then compute the residuals and SSR for your fitted line.

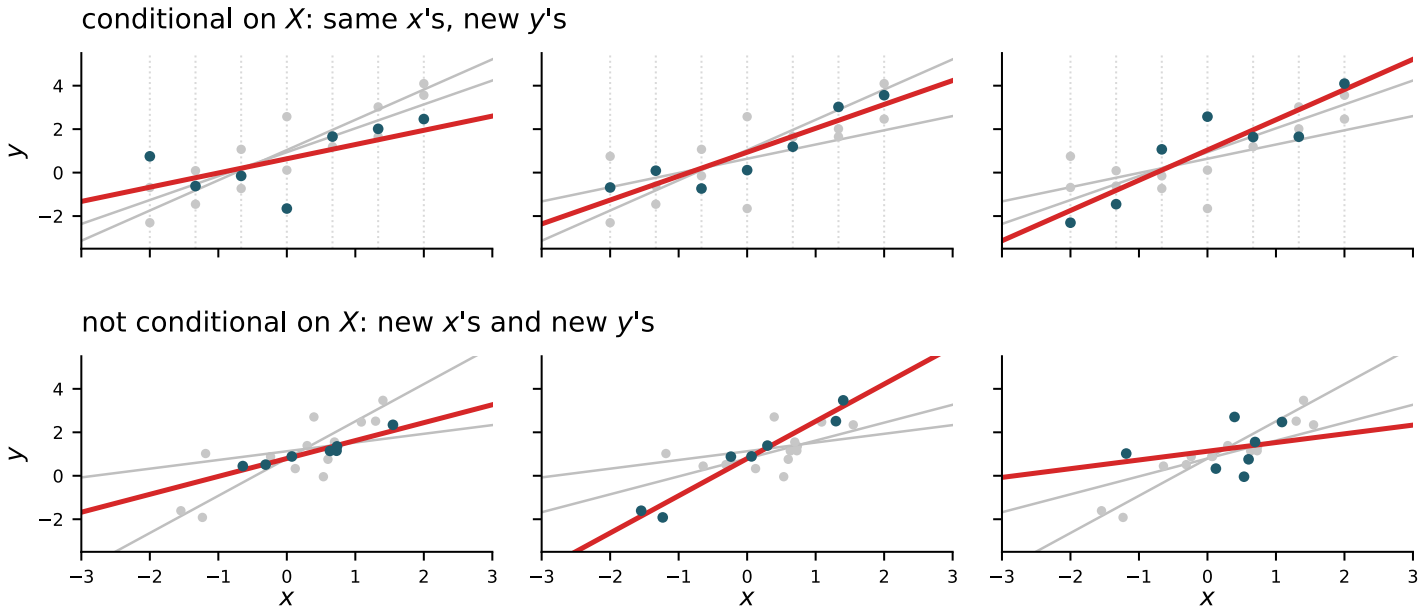
Outlier sensitivity

Now add a new data point $(X_6, Y_6) = (100, 3)$ to the dataset above. Explain, without doing a full calculation, why the fitted slope should change a lot. Then recompute the slope by hand and compare it to the original.

Sample distribution of the estimator

$\hat{\beta}_0$ and $\hat{\beta}_1$ are computed from random data, so they are themselves random variables: they change from replicate to replicate. This is the same idea as the sample distribution of $\overline{Y}$ from Section 3.2, now applied to the regression coefficients. For regression, though, there are two ways to make a replicate:

- **Conditional on X:** keep the same $X_1, \ldots, X_N$ and draw new $Y_i \mid X_i$. For example, an experiment where we choose the doses X_i ourselves.
- **Not conditional on X:** draw new pairs (X_i, Y_i), so the X's change too. For example, a survey, where a new sample brings new people with new X's.



Each panel highlights one replicate's data and least squares line; the other replicates in that row are in gray. Top: conditional on X (same x's, new y's). Bottom: not conditional on

X (new x's and new y's).

Which one makes sense depends on the context. We will focus on the sample distribution **conditional on** X, treating $X_1, \dots, X_N$ as fixed numbers; the spread of the X's then enters through

$$S_{xx} = \sum_{i=1}^{N} (X_i - \overline{X})^2 \approx N\sigma_X^2.$$

Example (unbiasedness of the slope estimator)

Question: show that $\hat{\beta}_1$ is unbiased, i.e. $E[\hat{\beta}_1 \mid X_1, \dots, X_N] = \beta_1$.

Solution: conditional on the X's, only the Y_i are random, so by linearity

$$E[\hat{\beta}_1 \mid X_1, \dots, X_N] = \frac{\sum_{i=1}^{N} (X_i - \overline{X})\, E[Y_i - \overline{Y} \mid X_1, \dots, X_N]}{\sum_{i=1}^{N} (X_i - \overline{X})^2}.$$

From the model, $E[Y_i \mid X_i] = \beta_0 + \beta_1 X_i$, and averaging over i gives $E[\overline{Y} \mid X_1, \dots, X_N] = \beta_0 + \beta_1 \overline{X}$, so

$$E[Y_i - \overline{Y} \mid X_1, \dots, X_N] = \beta_0 + \beta_1 X_i - \beta_0 - \beta_1 \overline{X} = \beta_1 (X_i - \overline{X}).$$

Therefore

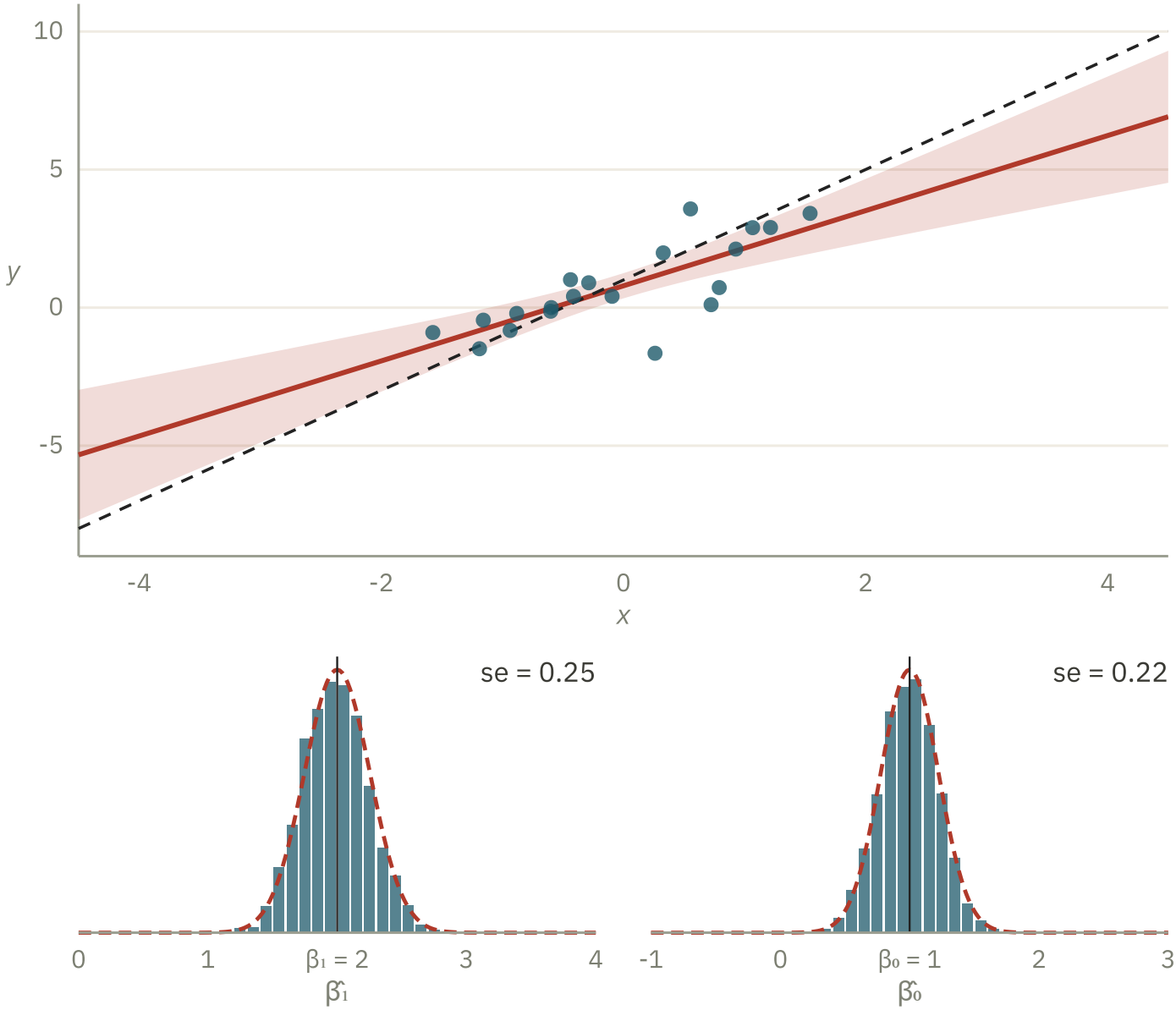
$$E[\hat{\beta}_1 \mid X_1, \dots, X_N] = \beta_1 \frac{\sum_{i=1}^{N} (X_i - \overline{X})^2}{\sum_{i=1}^{N} (X_i - \overline{X})^2} = \beta_1.$$

Across replicates with the same X's, $\hat{\beta}_1$ is correct *on average*. (A similar argument shows $E[\hat{\beta}_0 \mid X_1, \dots, X_N] = \beta_0$.)

One can also show (we won't derive this) that $\mathrm{var}(\hat{\beta}_1 \mid X_1, \dots, X_N) = \sigma_\epsilon^2 / S_{xx}$. Since $\hat{\beta}_1$ is a weighted sum of the Normal Y_i's, it is Normal, and using $S_{xx} \approx N\sigma_X^2$:

$$\boxed{\hat{\beta}_1 \approx \text{Normal}\left(\beta_1,\ \frac{\sigma_\epsilon^2}{N\sigma_X^2}\right)}, \qquad \text{se}(\hat{\beta}_1) \approx \frac{\sigma_\epsilon}{\sigma_X\sqrt{N}}.$$

The more spread out the X's (large σ_X), the less noise (small σ_ϵ), or the more data, the tighter the sample distribution of $\hat{\beta}_1$ is around β_1.



Top: one dataset from $Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$ with $\beta_0 = 1$, $\beta_1 = 2$, its fitted line (red), the 95% interval for the line (shaded) and the true line (dashed). Bottom: $\hat{\beta}_1$ (left) and $\hat{\beta}_0$ (right) over 2000 replicates with the same x's, with the Normal distributions $\text{Normal}(\beta_1, \sigma_\epsilon^2/S_{xx})$ and $\text{Normal}\big(\beta_0, \sigma_\epsilon^2(1/N + \overline{X}^2/S_{xx})\big)$ (dashed). Move μ_X, the mean of the x's, away from 0 and watch what happens to $\hat{\beta}_0$.

Drill

The sample distribution of $\hat{\beta}_0$

In this drill you will find the sample distribution of the intercept estimator $\hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X}$, conditional on $X_1, \dots, X_N$, starting from what we know about $\hat{\beta}_1$: it is Normal with $E[\hat{\beta}_1 \mid X_1, \dots, X_N] = \beta_1$ and $\text{var}(\hat{\beta}_1 \mid X_1, \dots, X_N) = \sigma_\epsilon^2 / S_{xx}$.

a. Find $E[\overline{Y} \mid X_1, \dots, X_N]$ and $\text{var}(\overline{Y} \mid X_1, \dots, X_N)$.

b. Use (a) and the unbiasedness of $\hat{\beta}_1$ to show that $\hat{\beta}_0$ is unbiased.

c. Conditional on the X's, $\overline{Y}$ and $\hat{\beta}_1$ are uncorrelated (you may use this without proof). Use it to show that

$$\text{var}(\hat{\beta}_0 \mid X_1, \dots, X_N) = \sigma_\epsilon^2 \left(\frac{1}{N} + \frac{\overline{X}^2}{S_{xx}} \right).$$

d. Using $S_{xx} \approx N \sigma_X^2$, write down the approximate sample distribution of $\hat{\beta}_0$ and its standard error.

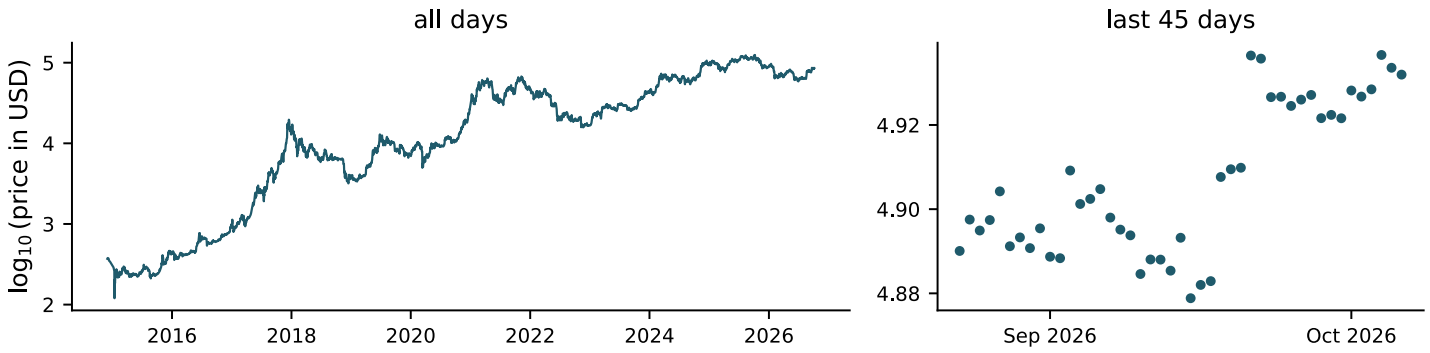
e. For which value of $\overline{X}$ is $\text{se}(\hat{\beta}_0)$ smallest, and what does $\hat{\beta}_0$ reduce to in that case? Explain with a sketch why the intercept is more uncertain when the X's are far from 0.

f. Optional: show that $\overline{Y}$ and $\hat{\beta}_1$ are uncorrelated conditional on the X's, using $\sum_i (X_i - \overline{X}) = 0$.

3.5 Autoregressive models and the Hurwicz bias

A time series $Y_1, Y_2, \dots, Y_N$ is a sequence of measurements of the same quantity at successive times. Such series are often modeled with regression models where the previous value plays the role of the predictor, and the next the response variable. The main difference between the previous examples is that now the data points $(Y_1, Y_2), (Y_2, Y_3) \dots$

are not independent. This is called an **autoregressive model**. We will use the example of the log price of bitcoin each day since December 2014 $Y_t = \log_{10}(\text{price on day } t)$.



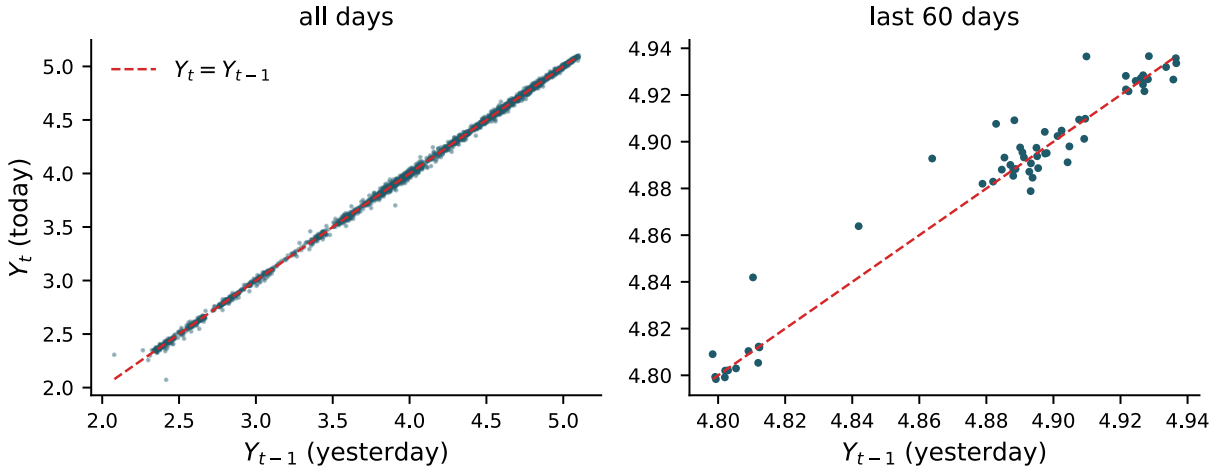
Log price of bitcoin vs. time. Left: all days. Right: the last 45 days.

The AR(1) model

The simplest autoregressive model, called AR(1), is

$$Y_t \mid Y_{t-1} \sim \text{Normal}(\beta_0 + \beta_1 Y_{t-1},\ \sigma_\epsilon^2), \qquad \text{i.e.} \qquad Y_t = \beta_0 + \beta_1 Y_{t-1} + \epsilon_t,$$

where the noise $\epsilon_t \sim \text{Normal}(0, \sigma_\epsilon^2)$ is independent of everything that happened before time t (but not he future). In particular, once we know Y_{t-1}, the earlier values $Y_{t-2}, Y_{t-3}, \cdots$ tell us nothing more about Y_t. This is a natural starting point for modeling many noisy processes, such as prices or height along a lineage (mother, daughter, granddaughter).



Today's log price vs. yesterday's. The dashed line is $Y_t = Y_{t-1}$.

For bitcoin, least squares gives $\hat{\beta}_1 \approx 0.9996$ and $\hat{\sigma}_\epsilon \approx 0.016$.

Stationary AR(1) processes

A series is **stationary** if its distribution does not change over time; in particular Y_t and Y_{t-1} have the same mean μ and the same standard deviation σ_Y. Three useful facts follow.

The slope is a correlation. Recall that the slope of any regression model is $\beta_1 = \mathrm{cov}(X, Y)/\sigma_X^2$. Here $X = Y_{t-1}$ and $Y = Y_t$ have the same standard deviation, so

$$\beta_1 = \frac{\mathrm{cov}(Y_{t-1}, Y_t)}{\sigma_Y^2} = \frac{\mathrm{cov}(Y_{t-1}, Y_t)}{\sigma_Y \, \sigma_Y} = \rho,$$

the correlation between consecutive values. For this reason we write ρ in place of β_1 from now on.

The mean. Taking the expectation of both sides of $Y_t = \beta_0 + \rho Y_{t-1} + \epsilon_t$ gives $\mu = \beta_0 + \rho\mu$, so $\beta_0 = (1-\rho)\mu$ and we can write the model as

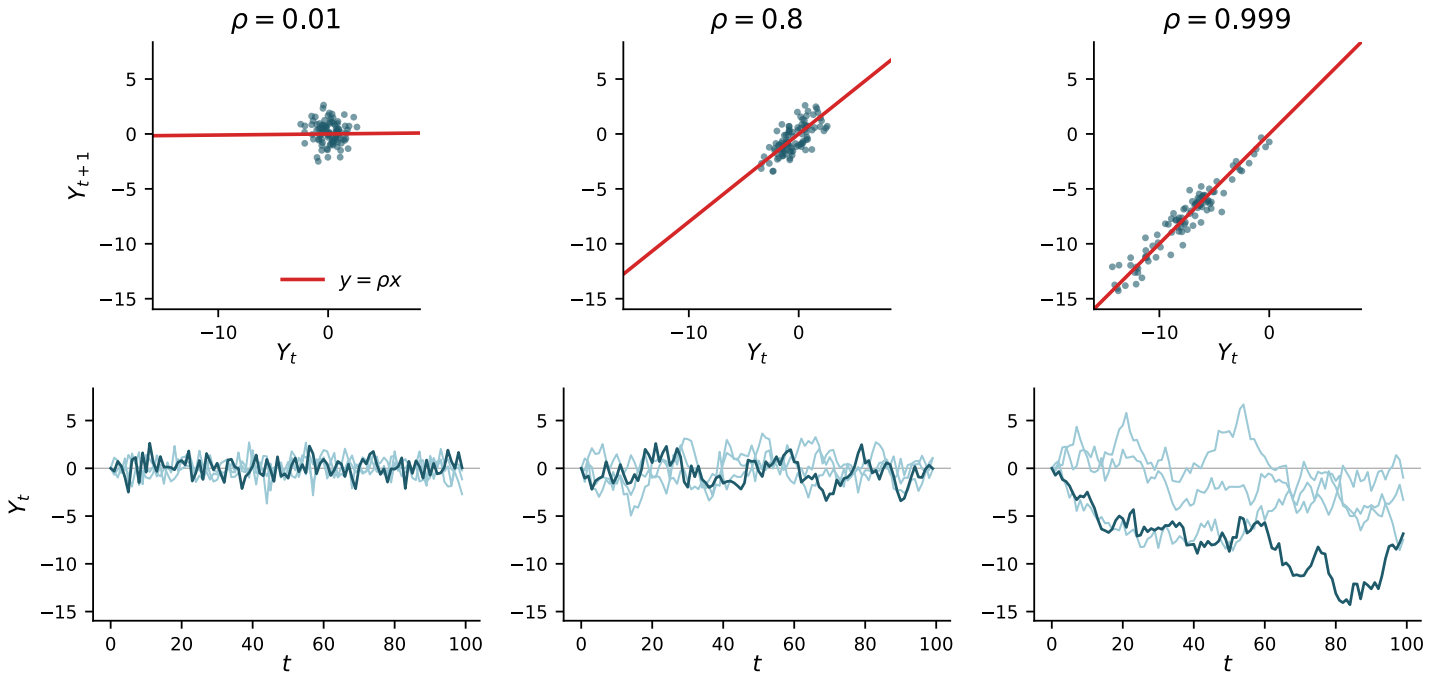
$$\boxed{Y_t - \mu = \rho\,(Y_{t-1} - \mu) + \epsilon_t.}$$

When $|\rho| < 1$, each step shrinks the deviation from the mean by a factor of ρ on average, and then adds new noise. This is **regression to the mean**: the daughters of unusually tall mothers are tall, but on average less tall than their mothers.

The variance. Taking the variance of both sides in the same way (and using that ϵ_t is independent of Y_{t-1}) gives $\sigma_Y^2 = \rho^2\sigma_Y^2 + \sigma_\epsilon^2$, so

$$\boxed{\sigma_Y^2 = \frac{\sigma_\epsilon^2}{1-\rho^2}.}$$

The closer ρ is to 1, the more slowly the series returns to its mean and the farther it deviates.



Simulated series $Y_{t+1} = \rho Y_t + \epsilon_{t+1}$ with $\epsilon \sim \text{Normal}(0, 1)$. Top: Y_{t+1} vs. Y_t for one series. Bottom: four series of 100 steps.

The random walk

What happens when $\rho = 1$? The formula above gives $\sigma_Y^2 = \infty$, so there is no stationary distribution. To see why, take $\beta_0 = 0$ and $Y_0 = 0$ and unroll the model:

$$Y_t = Y_{t-1} + \epsilon_t = Y_{t-2} + \epsilon_{t-1} + \epsilon_t = \cdots = \sum_{s=1}^{t} \epsilon_s, \qquad \text{so} \qquad Y_t \sim \text{Normal}(0,\ \sigma_\epsilon^2\, t).$$

This is a **random walk** (compare Problem 3.4): the noise is never forgotten, and the variance grows forever instead of settling down. With $\hat{\beta}_1 \approx 0.9996$, the log price of bitcoin is very close to a random walk.

Why least squares is biased: the Hurwicz bias

In Section 3.4 we saw that $\hat{\beta}_1$ is an unbiased estimator of β_1. For the AR(1) model this is no longer true. Take $\mu = 0$ for simplicity and fit $Y_t = \rho Y_{t-1} + \epsilon_t$ by least squares with no intercept. Substituting the model into the estimator,

$$\hat{\rho} = \frac{\sum_t Y_{t-1} Y_t}{\sum_t Y_{t-1}^2} = \frac{\sum_t Y_{t-1}(\rho Y_{t-1} + \epsilon_t)}{\sum_t Y_{t-1}^2} = \rho + \frac{\sum_t Y_{t-1}\,\epsilon_t}{\sum_t Y_{t-1}^2}.$$

In ordinary regression the noise ϵ_i is independent of *all* of the predictors $X_1, \dots, X_N$, so conditional on the X's the second term has mean 0 and the estimator is unbiased. **In a time series the samples are not independent**, and this argument breaks down: the predictors are $Y_1, \dots, Y_{N-1}$ themselves, and although ϵ_t is independent of the past value Y_{t-1}, it is part of Y_t, Y_{t+1}, and every later value. The noise is therefore correlated with the predictors that come after it, and the second term no longer averages to 0. The example below shows the result.

Example

Question: Simulate many AR(1) series with true $\beta_1 = 0.6$ (and $\beta_0 = 0$), and estimate β_1 by least squares (regressing Y_t on Y_{t-1}, no intercept) in each. Does the average of these estimates match 0.6? What happens as the series gets longer (larger N, the number of time points)?

Solution:

```
import numpy as np

rng = np.random.default_rng(42)
beta1 = 0.6

def fit_beta1_hat(N):
    eps = rng.normal(size=N)
    Y = np.empty(N)
    Y[0] = eps[0]
    for t in range(1, N):
        Y[t] = beta1 * Y[t - 1] + eps[t]
    y_lag, y = Y[:-1], Y[1:]
    return np.sum(y_lag * y) / np.sum(y_lag**2)

for N in [10, 50, 200]:
    beta1_hats = [fit_beta1_hat(N) for _ in range(20_000)]
    print(f"N={N}: average beta1_hat = {np.mean(beta1_hats):.3f}")
```

which prints

N	average $\hat{\beta}_1$
10	0.506
50	0.577
200	0.595

At every N the average estimate falls short of the true $\beta_1 = 0.6$, and the gap shrinks as N grows: $\hat{\beta}_1$ is biased downward in any finite sample, but consistent.

The size of the bias is harder to work out. For the no-intercept AR(1) model, it can be shown (Hurwicz, 1950) that for large N,

$$E[\hat{\beta}_1] - \beta_1 \approx -\frac{2\beta_1}{N}.$$

This matches the simulation: with $\beta_1 = 0.6$, the formula predicts a bias of about -0.024 at $N = 50$ and -0.006 at $N = 200$, both close to what we observed above (the match is worse at $N = 10$, since the formula is only a leading-order approximation for large N). The bias vanishes as $N \to \infty$, consistent with $\hat{\beta}_1$ being a consistent estimator, but at any finite N it pulls the estimate toward 0 — least squares systematically underestimates how persistent the process really is. This finite-sample downward bias of the AR(1) least-squares estimator is known as the **Hurwicz bias**. You do not need to memorize the formula; what matters is the reason the bias exists.

Autocorrelation

How long does a time series remember its past? The **autocorrelation** at lag k is the correlation between values k steps apart,

$$\rho_k = \operatorname{corr}(Y_t, Y_{t+k}) = \frac{\operatorname{cov}(Y_t, Y_{t+k})}{\sigma_Y^2},$$

where we used that Y_t and Y_{t+k} have the same standard deviation in a stationary series. We already know $\rho_1 = \rho$. For larger lags, applying the model k times writes Y_{t+k} as $\rho^k Y_t$ plus noise from times $t+1, \dots, t+k$, which is independent of Y_t (the drill below works through $k = 2$). Therefore

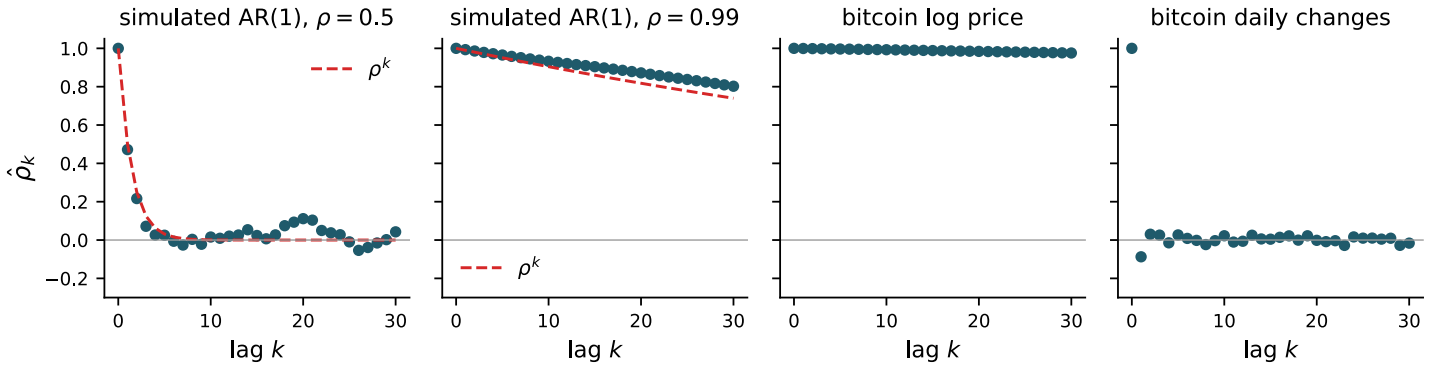
$$\boxed{\rho_k = \rho^k.}$$

The correlation decays *exponentially* in the lag. After $k = 10$ steps it is $0.5^{10} \approx 0.001$ when $\rho = 0.5$, but $0.9^{10} \approx 0.35$ when $\rho = 0.9$ and $0.99^{10} \approx 0.90$ when $\rho = 0.99$.

From data we estimate ρ_k with the sample version

$$\hat{\rho}_k = \frac{\sum_t (Y_t - \overline{Y})(Y_{t+k} - \overline{Y})}{\sum_t (Y_t - \overline{Y})^2},$$

and a plot of $\hat{\rho}_k$ vs. k is called the autocorrelation function.



Sample autocorrelation $\hat{\rho}_k$ vs. lag k for two simulated AR(1) series (dashed: ρ^k), the bitcoin log price, and its daily changes $Y_t - Y_{t-1}$.

For the simulated series, $\hat{\rho}_k$ follows ρ^k. The bitcoin log price stays correlated with itself for a very long time, as we expect from a series with $\rho \approx 1$, while its daily changes $Y_t - Y_{t-1}$ are almost uncorrelated: knowing how the price moved today tells us very little about how it will move tomorrow.

Drill

The stationary distribution of an AR(1) model

Consider the AR(1) model $Y_t = \beta_0 + \rho Y_{t-1} + \epsilon_t$ with $\beta_0 = 1$, $\rho = 0.7$ and $\sigma_\epsilon^2 = 4$.

a. Simulate one series of 2000 steps starting far from the mean, at $Y_0 = 50$, and plot Y_t vs. t. About how many steps does it take for the series to forget where it started?

b. Discard the first 500 steps as "burn-in," and compare a histogram of the remaining values to the Normal distribution with the stationary mean and variance from this section.

c. Repeat (a) and (b) starting from $Y_0 = -50$. Does the stationary distribution depend on Y_0?

d. Why do we need $|\rho| < 1$ for a stationary distribution to exist? Repeat (a) with $\rho = 1$ and with $\rho = 1.02$ and describe what you see.

e. Repeat (a) with $\rho = -0.7$. Is this series stationary? How does it look different from the series with $\rho = 0.7$?

The autocorrelation at lag 2

Let $Y_{t+1} = \rho Y_t + \epsilon_{t+1}$ be a stationary AR(1) series with mean 0. Compute $\text{cov}(Y_t, Y_{t+2})$ in terms of ρ and σ_Y^2. (The slides from Lecture 11 will help.)

Mothers and daughters

Suppose the heights of women along a lineage (mother, daughter, granddaughter, ...) follow a stationary AR(1) model, where Y_t is the height in inches of the woman in generation t. In the population the mean height is $\mu = 64$, the standard deviation is $\sigma_Y = 2.5$, and the correlation between the heights of a mother and her daughter is $\rho = 0.5$.

a. Find β_0 and β_1 in $Y_t = \beta_0 + \beta_1 Y_{t-1} + \epsilon_t$.

b. Find σ_ϵ.

c. A mother is 70 inches tall. What is the expected height of her daughter? Of her granddaughter?

Forecasting with a fitted AR(1) model

A meteorologist fits an AR(1) model to a long series of daily temperature anomalies Y_t (the day's temperature minus the usual temperature for that date, in °F) and finds $\hat{\beta}_0 = 0$, $\hat{\beta}_1 = 0.7$ and $\hat{\sigma}_\epsilon = 3$. Treat these estimates as the true values.

Today's anomaly is $+10$. What is the predicted anomaly tomorrow? Give an interval that contains tomorrow's anomaly with probability 95%.

Why least squares is biased for AR(1)

Consider the AR(1) model $Y_t = \rho Y_{t-1} + \epsilon_t$ with $0 < \rho < 1$.

a. For each pair, say whether the two random variables are independent: ϵ_t and Y_{t-1}; ϵ_t and Y_t; ϵ_t and Y_{t+3}.

b. In Section 3.4 we showed that $\hat{\beta}_1$ is unbiased conditional on $X_1, \ldots, X_N$. Explain in a sentence or two why that argument does not work when the predictors are $Y_1, \ldots, Y_{N-1}$.

c. Using the approximation $E[\hat{\rho}] - \rho \approx -2\rho/N$ with $\rho = 0.9$, how long a series do we need for the bias to be smaller than 0.01 in absolute value? Is $\hat{\rho}$ consistent?

d. Modify the code in the example above to check your answer to (c) by simulation.

3.6 Hypothesis testing

In statistics we often infer parameters not because we care about their exact values, but because we want to use them to make a decision — e.g. in a clinical trial, whether a

candidate drug is worth pursuing. This is often framed as **hypothesis testing**: we assign a probability to a hypothesis (or its converse). In abstract terms, the basic procedure is:

1. Come up with a **null hypothesis** — e.g. that the mean of some variable is zero — that we'd like to potentially rule out.
2. Compute a **test statistic** $\hat{T}$: like any estimator, some quantity computed from the data.
3. Ask a probabilistic question: what's the chance of observing a value of $\hat{T}$ at least as large as what we measured, *if the null hypothesis were true*? The result is the **p-value**.

Example (hypothesis testing for a binary-predictor regression model)

Consider a clinical trial again: an effect Y (e.g. blood pressure) is measured in a control group ($X = 0$) and a treatment group ($X = 1$), with $N/2$ people in each group. Model

$$Y \mid X \sim \text{Normal}\big(\mu_C(1 - X) + \mu_T X,\ \sigma_\epsilon^2\big),$$

assuming σ_ϵ^2 is known (this simplifies things a lot). This is a linear regression model, since

$$\mu_C(1 - X) + \mu_T X = \mu_C + (\mu_T - \mu_C)X = \beta_0 + \beta_1 X, \qquad \beta_0 = \mu_C,\ \ \beta_1 = \mu_T - \mu_C$$

We could estimate β_0, β_1 as usual, or equivalently estimate μ_C, μ_T directly within each group. Since σ_ϵ is known, the sample distributions are

$$\hat{\mu}_C \sim \text{Normal}\left(\mu_C,\ \frac{\sigma_\epsilon^2}{N/2}\right), \qquad \hat{\mu}_T \sim \text{Normal}\left(\mu_T,\ \frac{\sigma_\epsilon^2}{N/2}\right),$$

so

$$\hat{\beta}_1 \sim \text{Normal}\left(\beta_1,\ \frac{4\sigma_\epsilon^2}{N}\right).$$

Our null hypothesis is $\beta_1 = 0$ (no drug effect). As test statistic, we measure how far $\hat{\beta}_1$ is from zero in standard errors:

$$\hat{T} = \frac{\hat{\beta}_1}{\text{se}(\hat{\beta}_1)}.$$

Since σ_ϵ is known, $\text{se}(\hat{\beta}_1)$ is known too, so from the sample distribution's perspective this is just division by a constant.

Let $\hat{\beta}_1^*$ denote the measured effect *under the null hypothesis* — a replicate generated assuming $\beta_1 = 0$. Then $\hat{\beta}_1^*$ has the sample distribution shifted to be centered at zero:

$$\hat{\beta}_1^* \sim \text{Normal}\Big(0,\ \frac{4\sigma_\epsilon^2}{N}\Big).$$

We can now answer the question posed in step 3: if the null hypothesis were true, how likely would we be to see a $|\hat{T}|$ at least as large as what we observed? This is the p-value,

$$p_v = P\big(|\hat{T}^*| > |\hat{T}| \mid \hat{T}\big),$$

where $\hat{T}^*$ is the test statistic computed from $\hat{\beta}_1^*$ (probability taken over its distribution), while $\hat{T}$ is fixed by our actual data.

The demo below shows both sides of this at once. On the left is the null distribution of $\hat{\beta}_1^*$ with the two-sided p-value shaded — the chance of seeing something at least as extreme as $\hat{\beta}_1$. On the right is the 95% confidence interval for $\hat{\beta}_1$, drawn against 0.